Lecture Notes: Finance—Spreadsheets; Simple Interest

Math F113X Math and Society

Basic Computation

Getting started with spreadsheets.

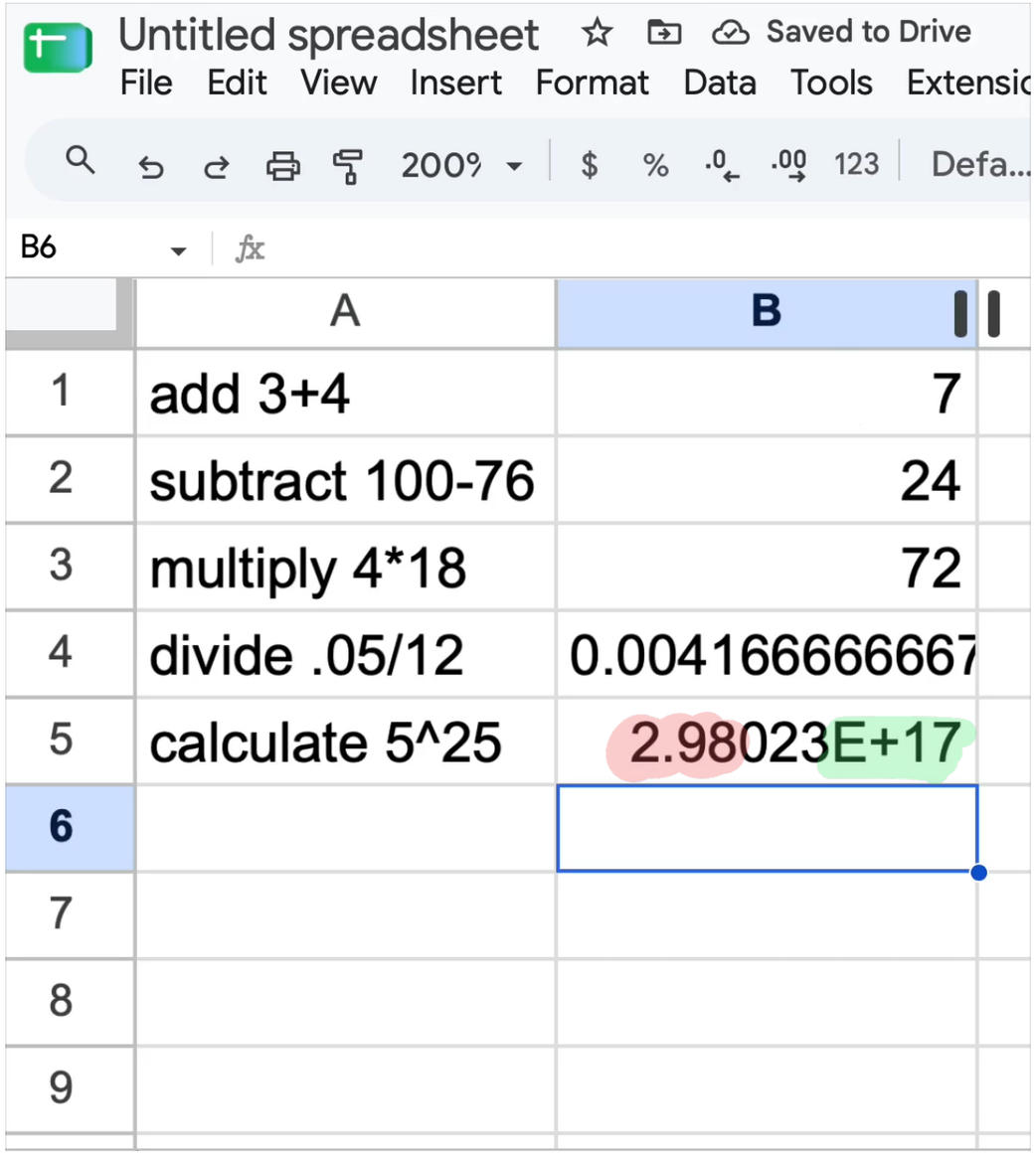
(a) To add 3 + 4, enter `=3+4`

(b) To subtract 100-76, enter `=100−76`

(c) To multiply 4 times 18, enter `=4*18`

(d) To divide 0.05 by 12, enter `=0.05/12`

(e) To calculate 5^{25}, enter `=5^25`



	A	B
1	add 3+4	7
2	subtract 100-76	24
3	multiply 4*18	72
4	divide .05/12	0.004166666667
5	calculate 5^25	2.98023E+17
6		
7		
8		
9		

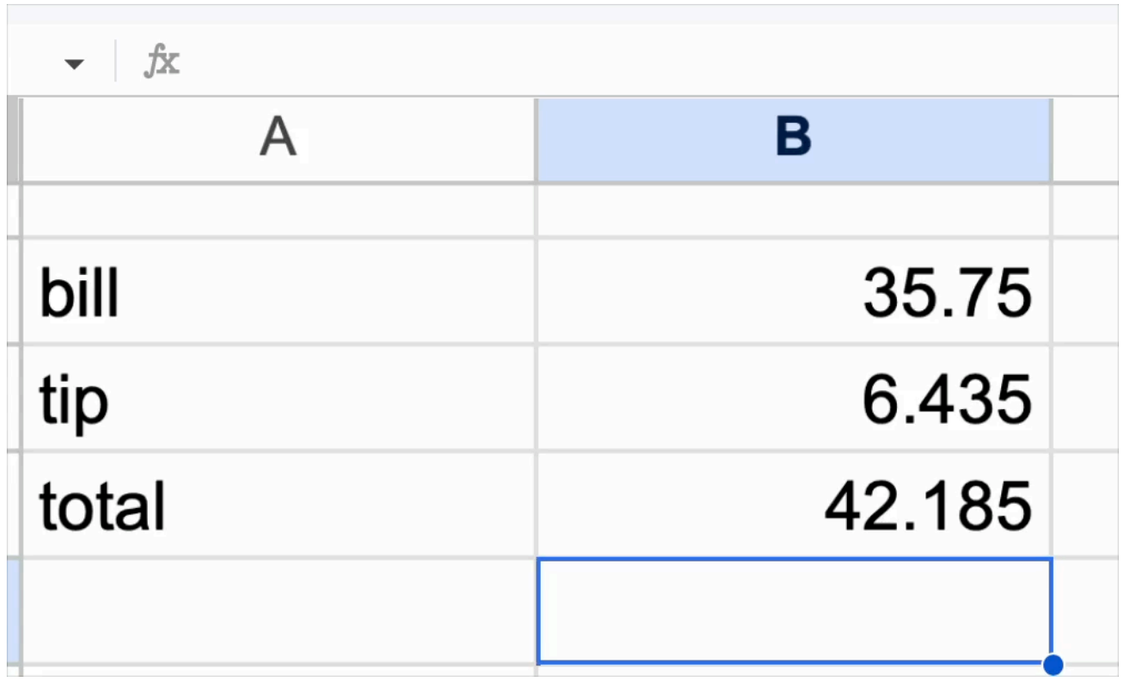
Compute an 18% Tip on a $35.75 Bill

(a) What is 18% as a decimal? $18\% = \frac{18}{100} = 0.18$

percent = per centum
= per hundred

(b) What should we enter into the spreadsheet?

Compute
$= 35.75 * (0.18)$

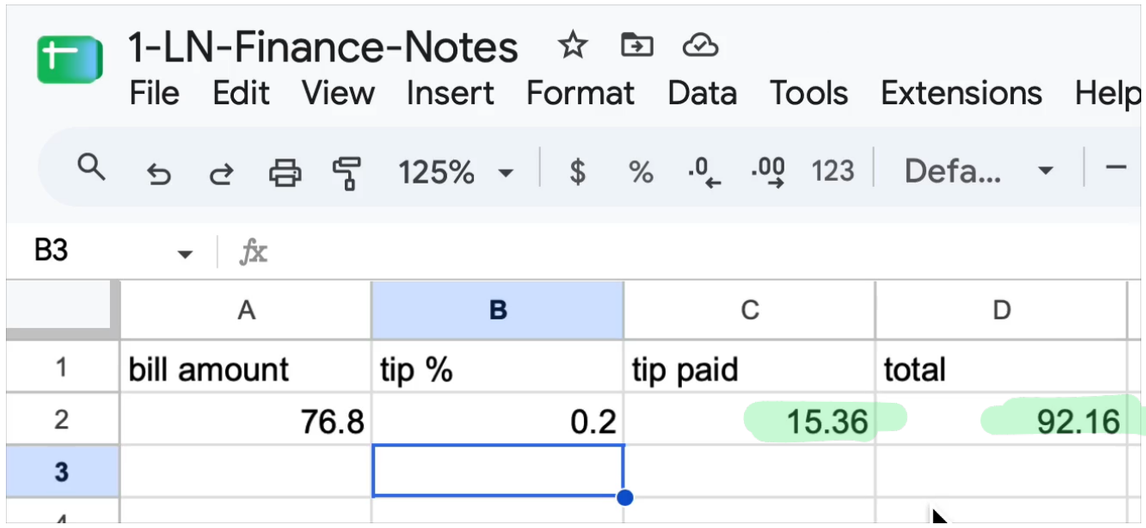


Make a Tip Calculator Using **Cell References**

(a) Open a new spreadsheet/tab and label it `TIP`.

(b) Fill in the spreadsheet:

	A	B	C	D
1	Bill Amount	Tip Percent	Tip Paid	Total
2	=35.75	=0.18	=A2 * B2	=A2 + C2



(d) What calculation did we do?

(e) Use the calculator to determine the Total if the Bill Amount is \$76.80 and the Tip Percent is 20%.

The **Fill Down** Feature

Keep tip at 20%, modifying our previous spreadsheet

(a) Enter 10 into cell A2 and enter =A2+10 into cell A3

(b) Select A3 and drag down until you get to 110.

(c) Select cells B2 through D2 and drag the blue dot in the right-hand corner down.

(d) How much is the tip on a \$110 meal?

\$22

How much is the final bill? \$132

How large does the Bill Amount need to be for the tip to be more than \$15?



	A	B	C	D
1	bill amount	tip %	tip paid	total
2	\$10.00	0.2	\$2.00	\$12.00
3	\$20.00	0.2	\$4.00	\$24.00
4	\$30.00	0.2	\$6.00	\$36.00
5	\$40.00	0.2	\$8.00	\$48.00
6	\$50.00	0.2	\$10.00	\$60.00
7	\$60.00	0.2	\$12.00	\$72.00
8	\$70.00	0.2	\$14.00	\$84.00
9	\$80.00	0.2	\$16.00	\$96.00
10	\$90.00	0.2	\$18.00	\$108.00
11	\$100.00	0.2	\$20.00	\$120.00
12	\$110.00	0.2	\$22.00	\$132.00
13	\$120.00			
14	\$130.00			
15				

Quick Summary of Finance Terms

(a) **principal** Starting amount of $ (deposit); present value

(b) **interest** an amount of money added in each time period

(c) **interest rate** r
a percent that determines how much money is added per time period

(d) **accumulated amount** or **end amount** or **principal plus interest** or **future value**
$A = P + I$ ⇝ simple interest means interest is calculated only on the principal!

(e) **Formula for Simple Interest over Time**
$A = P + I = P + P \cdot r \cdot t = P(1 + rt)$

(f) Comment on time t. (r: rate, t: time)
interest can be added over different time periods!

Make a Simple Interest Calculator

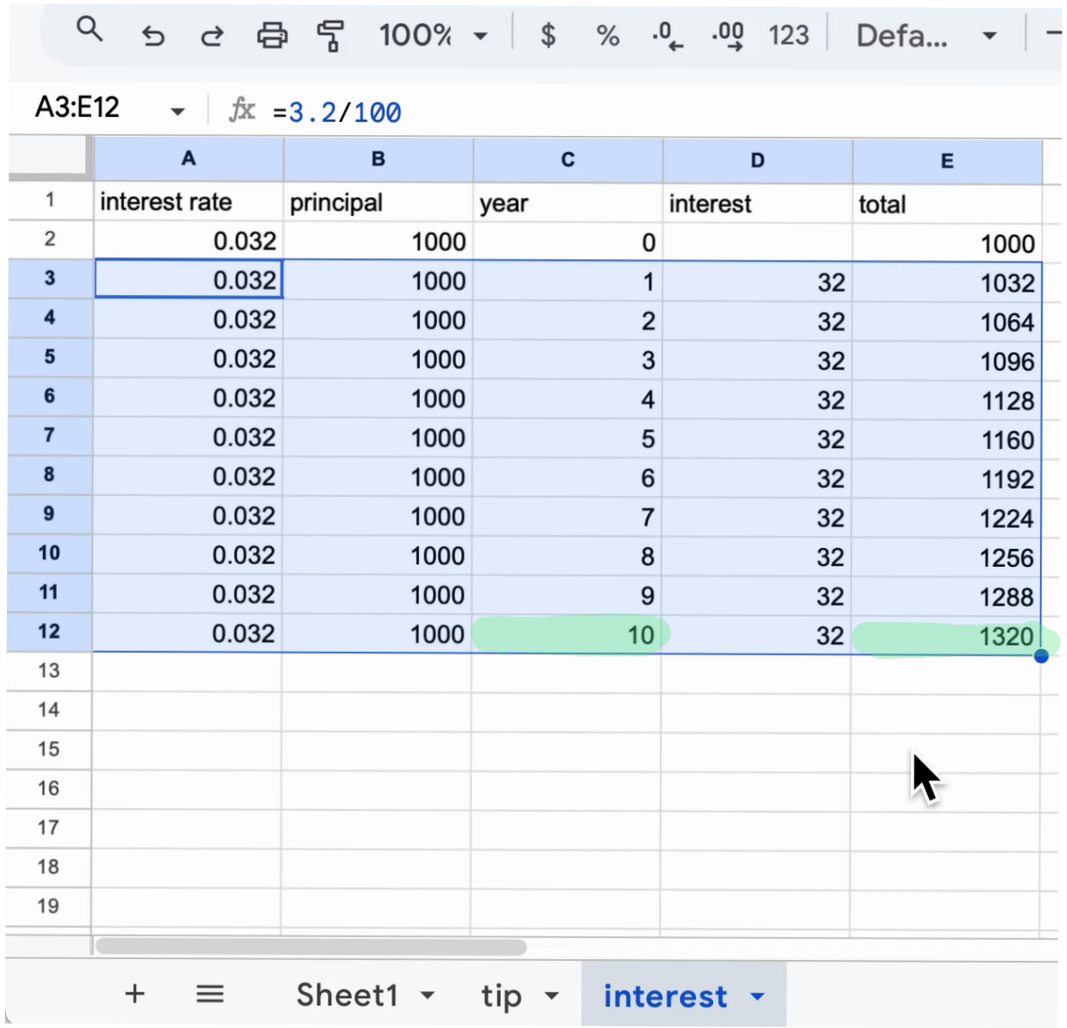
Suppose \$1000 is invested in a savings account that earns 3.2% interest on the \$1000 each year.

(a) Open a new spreadsheet/tab called `Interest`. Fill in the spreadsheet cells:

	A	B	C	D	E
1	Interest Rate (r)	Principal (P)	Year (t)	Interest (I)	Total (A)
2	=0.032	=1000	0		=B2+D2
3	=A2	=B2	=C2+1	=A2*B2	=D3+E2

Make a Simple Interest Calculator

(b) Use the pull down feature to determine the Total in the savings account in 10 years. Write down your steps.



	A	B	C	D	E
1	interest rate	principal	year	interest	total
2	0.032	1000	0		1000
3	0.032	1000	1	32	1032
4	0.032	1000	2	32	1064
5	0.032	1000	3	32	1096
6	0.032	1000	4	32	1128
7	0.032	1000	5	32	1160
8	0.032	1000	6	32	1192
9	0.032	1000	7	32	1224
10	0.032	1000	8	32	1256
11	0.032	1000	9	32	1288
12	0.032	1000	10	32	1320
13					
14					
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16					
17					
18					
19					

Make a Simple Interest Calculator

(c) How long will it take for the savings account to double in value? Write down your steps.

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125% $ % .0 .00 123 Defa... 10 B I

A34:E34 =D34+E33

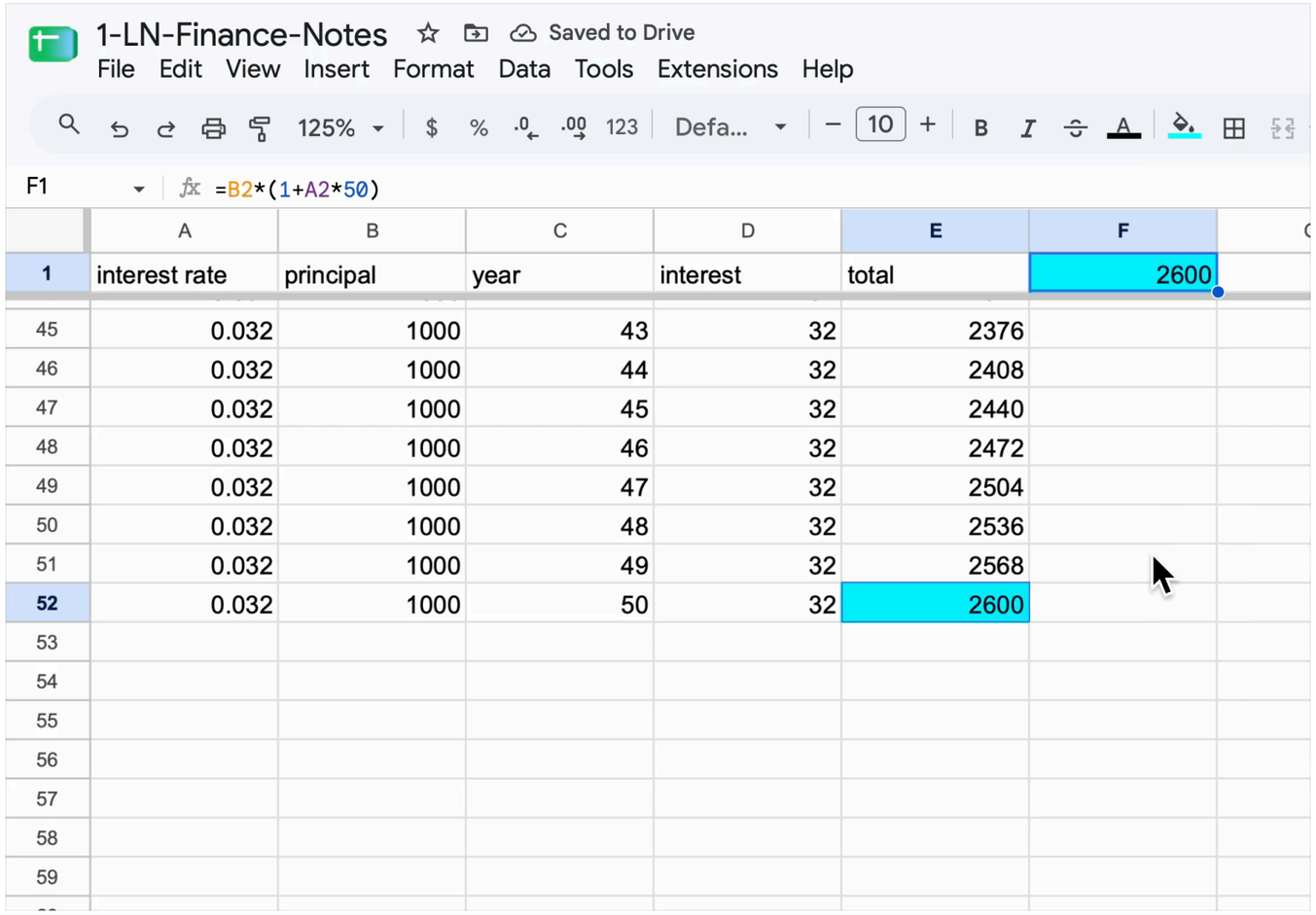
	A	B	C	D	E
1	interest rate	principal	year	interest	total
24	0.032	1000	22	32	1704
25	0.032	1000	23	32	1736
26	0.032	1000	24	32	1768
27	0.032	1000	25	32	1800
28	0.032	1000	26	32	1832
29	0.032	1000	27	32	1864
30	0.032	1000	28	32	1896
31	0.032	1000	29	32	1928
32	0.032	1000	30	32	1960
33	0.032	1000	31	32	1992
34	0.032	1000	32	32	2024
35	0.032	1000	33	32	2056
36	0.032	1000	34	32	2088
37	0.032	1000	35	32	2120
38	0.032	1000	36	32	2152

Make a Simple Interest Calculator

(d) How can you calculate the Total in the savings account in 50 years **without** using the pull down feature?

Write a formula for A. Then check your calculation using the pull down feature.

$$A = P(1+rt)$$
$$= 1000(1 + 0.032 * 50)$$



	A	B	C	D	E	F
1	interest rate	principal	year	interest	total	2600
45	0.032	1000	43	32	2376	
46	0.032	1000	44	32	2408	
47	0.032	1000	45	32	2440	
48	0.032	1000	46	32	2472	
49	0.032	1000	47	32	2504	
50	0.032	1000	48	32	2536	
51	0.032	1000	49	32	2568	
52	0.032	1000	50	32	2600	
53						
54						
55						
56						
57						
58						
59						

Make a Simple Interest Calculator

(e) A better simple interest calculator, using fixed cell references (the $ fixes the row/column)

	A	B	C	D	E
1	Interest Rate	Principal	Year	Interest	Total
2	0.032	1000	0		=B2
3			=C2+1	=A2*B2	=D3+E2

Now select and pull down C3, D3, E3. What happens if you change the interest rate to 4%? What if you change the principal to $3000?

Make a Simple Interest Calculator

What happens if you change the interest rate to 4%? What if you change the principal to $3000?

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125% $ % .0 .00 123 Defa... − 10 + B I A

B3 fx

	A	B	C	D	E	F
1	interest rate	principal	year	interest	total	
2	0.04	3000	0		3000	
3			1	120	3120	
4			2	120	3240	
5			3	120	3360	
6			4	120	3480	
7			5	120	3600	
8			6	120	3720	
9			7	120	3840	
10			8	120	3960	
11			9	120	4080	
12			10	120	4200	
13			11	120	4320	
14						
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+ ≡ Sheet1 tip interest better interest

How To Export the **Entire Workbook** as a **PDF**

(a) Save your work!

(b) In upper left menu, click `File` to produce a dropdown menu.

(c) In dropdown menu, select `Download`

(d) In right-side menu, select `PDF`